

Audio

In all scanning suites an audio system is in place to enable the presentation of an auditory stimulus to volunteers undergoing imaging.

The information below details the default audio installations available.

Headphone/Earphone Options

Please follow links below for information on the headphone systems supplied by the standard audio set-up.

Available Headphones				
Classification	Name	Type	Available	Restrictions
Standard	Etymotic	Earphone	3T, 7T, MEG	None
Standard	CONFON	Headphone	3T	Only fits within 20ch coil
Non-Standard	Sensimetric Audio	Earphone	3T	Requires training

MRI Audio

In the MRI Scanning suites, audio can be played to volunteers undergoing MRI through [various headphones](#). The auditory signal can also be record as an analogue data channel via the 1401 and Spike software.

The following shows the direction of the audio signal to volunteer.

STIM-PC sound generation → Windows system volume → Edirol UA-4FX USB soundcard → Audio Splitter → Headphone

MRI Instructions

1. Connect STIM-PC to audio output

1. In all instances follow the [STIM-PC instructions](#)
2. Connecting the 'Master USB' to the STIM-PC in use, will permit auditory presentation to a volunteer.

2. Set audio output volume levels

1. The audio volume level can be changed in various places. Researchers should test volume levels during technical pilots and ensure they use of their specific settings.

2. Default volume levels:

1. STIM-PC: *System volume 100%*
2. Edirol USB Soundcard: *Volume 100%*

3. Audio Splitter: *Volume mark 5*
4. A volume increase should be made on the relevant dial of the splitter.

3. **Test audio output**

1. Ensure audio is being playing as expected
2. Ensure volume levels are safe prior to testing on a volunteer

4. **Connect headphones to volunteer**

1. Follow the relevant headphone guides

5. **Audio check with volunteer**

1. Ensure the volume is suitable
2. Ensure left and right audio is heard

MRI Audio Schematic

See below the connection of the auditory system in MRI suites.

MEG Audio

In the MEG suite audio can be presented to volunteers via etymotic earphones and be recorded as an analogue data channel in the MEG data, via the MEG DAC.

If audio timing and synchronisation are not critical, the following method is recommended:

STIM-PC sound generation → Windows system volume → Audiofile soundcard → Audio Splitter → Etymotic Earphone

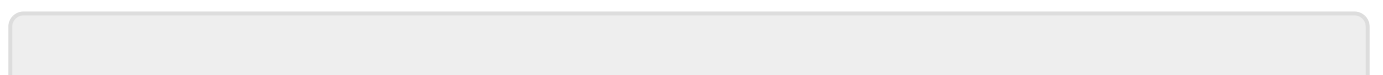
If audio timing and synchronisation are critical, the following method is recommended:

STIM-PC trigger → Audiofile sound generation, via buffered sounds on an SD card → Audio Splitter → Etymotic Earphone

MEG Instructions

MEG Audio Schematic

See below the connection of the auditory system in MEG.



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